

Lyrics as Mirrors: Tracing Chinese Social Shifts in Mandarin Popular Music from 2019 to 2024

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1. Introduction

Music is the heartbeat of its time, thrumming with the rhythm of the social climate. In China, Mandarin popular music, known as Mandopop, is a genre of Chinese music composed and performed in Mandarin that gains widespread popularity and commercial success (Lim & Kong, 2018; Hunke et al., 2025). It serves as a leading cultural influence, with lyrics that constantly evolve with the national narrative. Notably, the period from 2019 to 2024 has undergone a rapid societal transformation, accelerated by global events such as the COVID-19 pandemic and advances in digital technologies, which have rewritten youth identity and broader societal perceptions (Amankwah-Amoah et al., 2021; Holl et al., 2025). The study aims to investigate how lyrical topics and sentiments changed between 2019 and 2024 and to analyze how lyrical evolution reflects sociopolitical, sociocultural, and socioeconomic shifts in China.

The study is driven by three objectives as follows.

- (a) Identify key lyrical topics in Mandopop from 2019 to 2024.
- (b) Examine how sentiment trends in lyrics change from 2019 to 2024.
- (c) Explore how shifts in lyrical topics and sentiments mirror broader societal attitudes.

1.1 Significance

The study's significance is clear on two levels: theoretical and practical.

Theoretically, the study contributes to the emerging field of digital humanities. By applying Natural Language Processing (NLP) techniques to Mandopop lyrics, this approach goes beyond mere textual analysis to provide a data-driven, dynamic description of how China's rapid social transformation between 2019 and 2024 is encapsulated within its popular music. The study also extends the established quantitative and qualitative frameworks for music analysis proposed by Xia et al. (2019) and Hunke et al. (2025) by applying them to the unique linguistic and cultural context of contemporary China, with

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explicit focus on the complex collective sentiment expressed during this dynamic period.

Practically, the study can further address real-world issues. Remarkably, it challenges the notion that popular music is shallow, offering a deeper understanding of how popular songs interact with social trends in contemporary China. In particular, policymakers can use evolutionary trends in lyrical topics and sentiments as indicators of collective emotional states. Social researchers can likewise leverage these insights to explore emotional shifts during significant societal events, such as the COVID-19 outbreak, to better understand societal trauma and public recovery. For music practitioners, knowledge of these longitudinal trends in lyrical and sentiment content provides a basis for anticipating future market demand and informs strategic marketing campaigns to produce songs that best appeal to the general public.

2. Literature Review.

2.1 Song Lyrics as a Mirror of Social Changes.

Previous scholars worldwide have consistently demonstrated that popular music lyrics are a reliable indicator of societal transformation, reflecting values and sentiments. Pioneering research by Xia et al. (2019) introduced a robust quantitative framework based on the Latent Dirichlet Allocation (LDA) model for analyzing the evolution of pop music over time and its interrelationships with societal changes. Further affirming the validity and applicability of the NLP-based methodology, Hunke et al. (2025) conducted a parallel study on German popular music, demonstrating its utility across different cultural contexts. The study builds upon these foundations, adapting them to China's unique social landscape from 2019 to 2024.

2.2 Topic Modeling in Mandopop Lyrics.

The analysis of Mandopop lyrics necessitates specialized topic modeling techniques due to the linguistic complexities of Mandarin. Qin et al. (2016) showed that the Chinese Character-Word Topic Model (CWTM) significantly outperforms both word-based and character-based models for Chinese text. Thus, CWTM serves as the technical foundation of the analysis. The integration of a specialized Python word-segmentation library, Jieba, with topic models has been validated as a suitable means for Chinese text mining, as proposed by Sun Junyi and systematically demonstrated by Gao and Song et al. (Gao et al., 2022; Song & Yu, 2022; Zhang, 2024). Additionally, the study incorporates insights from the theme-aware generation model developed by Wang & Zhao (2019), which combines a Multi-Channel Seq2Seq architecture with Latent Dirichlet Allocation (LDA) to enhance topic coherence. For thematic categorization, the study draws upon the established broad categories of music topics from Hunke et al. (2025), adapting them to the study's context. These include: (1) Loneliness and Thinking, (2) Time and Memories, (3) Self Discovery, (4) Life and Society, (5) Dreams and Pursuit, and (6) Love and Emotion.

2.3 Sentiment Analysis in Chinese Lyrics.

The methodology for sentiment analysis of Chinese lyrics has evolved from lexicon-based approaches to sophisticated hybrid models. Previous research by Hu et al. (2009) proposed a lexicon-based method for detecting emotions in Chinese lyrics, laying a foundation for the field. A recent study by Wang and Wong (2025) developed a hybrid sentiment analysis framework that integrated large language models with a comprehensive Chinese emotion lexicon, outperforming the lexicon-only method proposed by Hu et al. (2009). This advanced framework thus provides essential methodological support for the study to accurately capture complex emotional tendencies in contemporary Chinese popular lyrics.

3. Methodology and Data.

The study employs NLP techniques, including text mining, topic modeling, and sentiment analysis, using Python libraries such as pandas, NumPy, scikit-learn, and Jieba for data processing. It also uses visualization tools, including Matplotlib, seaborn, and WordCloud.

First, text mining extracts the names and popularity scores of the weekly top 10 songs from the TME (Tencent Music Entertainment Group) Uni Chart. This data is then sorted by overall average score, with the top 100 songs per annum included. Next, the lyrics of these 600 songs underwent rigorous preprocessing: many variables (e.g., artist names, producer credits) were removed, standard and custom stop words (e.g., onomatopoeia and filler words like “呢” and “woo”) were filtered out, and text was tokenized and lemmatized to reduce lexical variance.

Topic modeling is then used to organize the overall topic distributions and word frequencies found in the processed data. Using Latent Dirichlet Allocation (LDA), we identify six distinct lyrical topics, previously stated. The words in the processed lyrics are analyzed for frequency and contribution to each topic, yielding insights into key trends and high-frequency topics per annum. Visualizations such as word clouds, pie charts, and trend graphs are created to illustrate the findings.

Finally, sentiment analysis, specifically tailored to Chinese linguistics, is applied to the processed lyrics. This approach generates trend graphs for four sentiments (i.e., Positive, Negative, Ambiguous, and Neutral) derived from the six identified topics, allowing us to observe the emotional trajectory of each topic over time. Notably, a dictionary-based lexicon approach is used to match lyrics with a predefined sentiment lexicon containing texts with assigned polarities. This enables researchers to better manually capture niche vocabulary and cultural meanings. (Mozetič, Grčar, & Smailović, 2016).

3.1 Data.

Our dataset comprises 3,120 top-ranked tracks from the TME Uni Chart over six years, covering the top 100 songs annually (600 in total), to identify musical trends. To elaborate on our data, TME Uni Chart, a music ranking list, stands out by integrating data from QQ Music, KuGou Music, and Kuwo Music to create a comprehensive operating system and exclusive access to a vast array of artists, including icons like Jay Chou, surpassing its competitors. The combined monthly active users (MAU) of TME's platforms, around 650 million, significantly exceed other market players, such as Netease Music. TME Uni Chart is likewise updated weekly and accurately reflects the popularity of songs in the Chinese market. Therefore, TME Uni Chart serves as a reliable source for identifying the top 100 songs each year, ranked by overall score.

4. Findings and Analysis.

4.1 Topic Modeling Overview.

Six topics were selected for their distinctiveness and ability to capture the main themes in the corpus, aligning with Hunke et al.'s (2025) music framework while highlighting the nuances of Chinese popular music from 2019 to 2024. These six Mandopop lyric topics, with associated words, songs, and probabilities, are as follows.

Table 1. Overview of identified topics (all values in %).

Topic No.	Associated words	Associated songs	Topic names
1	我会 (12.3), 遇见 (11.1), 世界 (9.7), 回答 (9.0), 回忆 (7.9), 看到 (7.6), 季节 (7.4), 时间 (7.2), 相信 (7.1), 再见 (7.0), 可能 (6.9), 青春 (6.8)	渐暖 (Getting Warmer) – Teens In Times (TNT) (98.93) 圣诞星 (Christmas Star) – Jay Chou & Cindy Yen (98.54) 烟花升停在星夜 (Starry Night) – Teens In Times (TNT) (98.42) 5:23 PM – Song Yaxuan (98.34) Falling You – Liu Yaowen (98.21)	Time and Memories
2	想念 (12.5), 心跳 (11.0), 跳舞 (9.8), 晚安 (9.0), 世界 (8.8), 拥抱 (8.3), 怀念 (7.7), 不会 (7.2), 天空 (6.5), 眼泪 (6.5), 鸟儿 (6.4), 等待 (6.2)	小美满 (Little Joys YOLO) – Zhou Shen (98.86) 听悲伤的情歌 (Listening to sad love song) – Su Xingjie (98.24) 像阳光那样 (Just Like Sunshine) – Wang Yibo (97.33) 睡个好觉 (Wake Back To Bed) – Zhou Shen (98.0) 不冬眠 – Liu Yaowen (98.0)	Love and Emotion
3	世界 (11.4), 江湖 (10.8), 希望 (10.6), 存在 (10.6), 囚禁 (9.2), 孤独 (7.7), 可爱 (7.7), 不负 (7.4), 起来 (6.3), 相遇 (6.2), 相信 (6.2), 记得 (5.8)	寻一个你 (To Find You) – Liu Yuning (98.58) 亢奋 (Euphoria) – Yan Haoxiang (98.5) Spotlight – Xiao Zhan (97.86) I'm A Star – Zhou Shen (97.53) 若想念飞行 (Missing Flying) – Teens In Times (TNT) (97.45)	Dreams and Pursuit
4	无人 (14.8), 完美 (12.3), 想念 (9.4), 还有 (8.1), 喜欢 (7.7), 小孩 (7.7), 世界 (7.3), 习惯 (6.9), 流浪 (6.5), 失去 (6.4), 生活 (6.4), 地方 (6.4)	抱我 (Hug Me) – Cai Xukung (99.1) 绝配 – Teens In Times (TNT) (99.09) No Complaints – Yan Haoxiang (97.76) 躺着真舒服 (Lying Down) – Teens In Times (TNT) (97.74) 离别开出花 (Parting blossoms) – Nan Fangkai (97.66)	Life and Society
5	期待 (13.8), 意义 (10.8), 世界 (9.6), 时间 (8.0), 离开 (7.6), 宇宙 (7.4), 梦见 (7.3), 告诉 (7.1), 想要 (7.1), 思念 (7.1), 不会 (7.1), 实现 (7.1)	最伟大的作品 (The Great Works) – Jay Chou (98.55) 偷 – Ma Jiaqi (98.49) 有你 (I Have You) – Zhao Lei (97.87) 百忧戒 – Teens In Times (TNT) (98.39) 若月亮没来 (If The Moon Does not Come) – Leto (97.81)	Self Discovery
6	不会 (13.2), 相遇 (11.9), 咖啡 (11.9), 是否 (9.1), 世界 (8.5), 才能 (7.4), 城市 (6.8), 还有 (6.5), 长大 (6.5), 幸福 (6.2), 奇妙 (6.1), 回到 (6.0)	璀璨冒险人 (The Wanderer) – Zhou Shen (97.56) 你要快乐 – Teens In Times (TNT) (98.11) 暮色回响 (Twilight Echo) – Angela Zhang (97.81) 爱到1440 – Teens In Times (TNT) (97.5) 姑娘在远方 – Ke Ke Ke A (97.48)	Loneliness and Thinking

Table 2. Overview of identified topics and sentiment.

Topic/sentiment	Positive		Negative		Ambiguous		Neutral	
	b	R²	b	R²	b	R²	b	R²
Time and Memories	-0.02	0.022	0.017	0.024	0.006	0.02	0.005	0.034
Love and Emotion	0.02	0.021	-0.027	0.012	0.109	0.017	-0.057	0.011
Dreams and Pursuit	-0.083	0.012	0.023	0.02	0.027	0.028	0.023	0.023
Life and Society	0.066	0.016	0.025	0.005	-0.051	0.008	-0.024	0.012
Self Discovery	-0.038	0.008	-0.008	0.024	-0.047	0.01	0.053	0.023
Loneliness and Thinking	0.03	0.019	-0.046	0.006	-0.073	0.024	0.038	0.02

4.2 Topic Distribution Across Years.

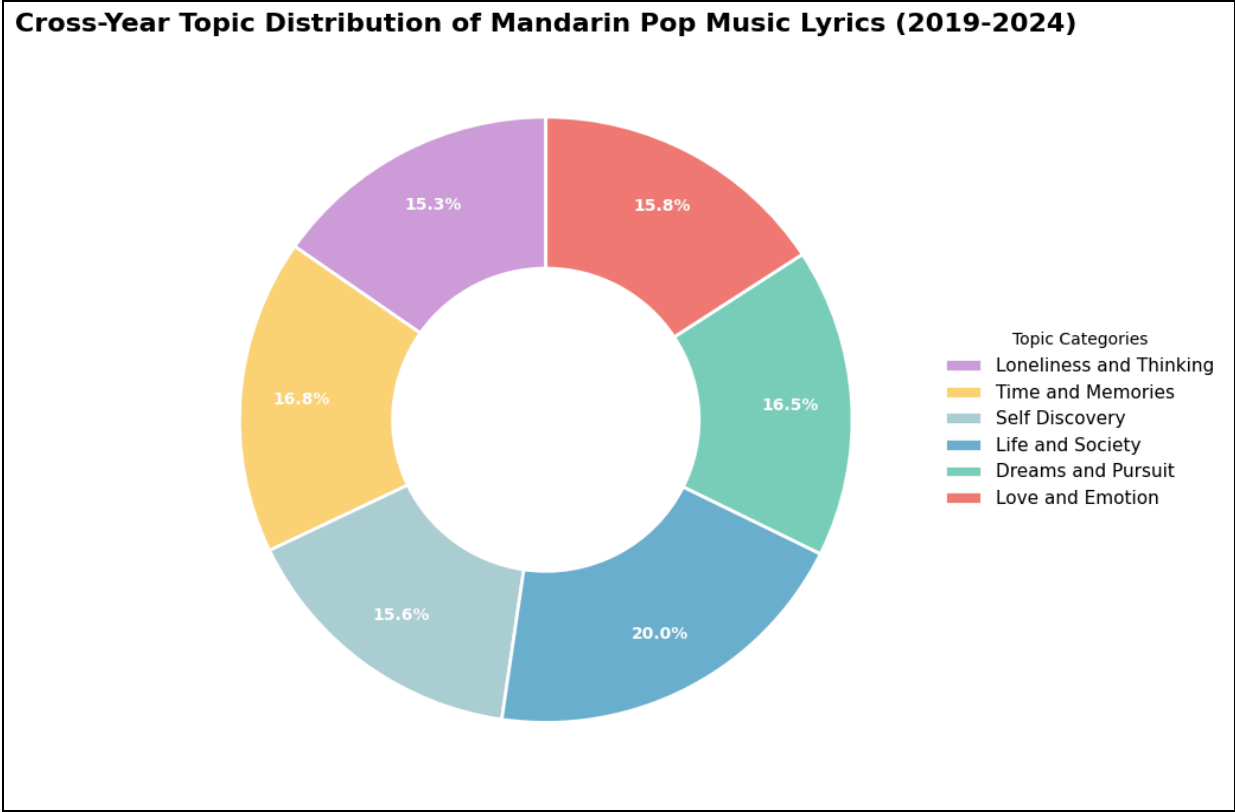


Figure 1: Cross-Year Topic Distribution of Mandarin Pop Music Lyrics 2019-2024

The overall distribution of lyrical topics in Mandopop from 2019 to 2024 provides a foundational insight into Mandopop’s thematic landscape (Figure 1). The chart suggests a balanced representation of topics, indicating a diverse exploration of human experience. Life and Society (20.0%) is the most common category, implying that Mandopop lyrics are primarily rooted in collective human issues. However, the overall balance indicates that the genre’s enduring appeal stems from its broad focus on connection, aspiration, and introspective commentary.

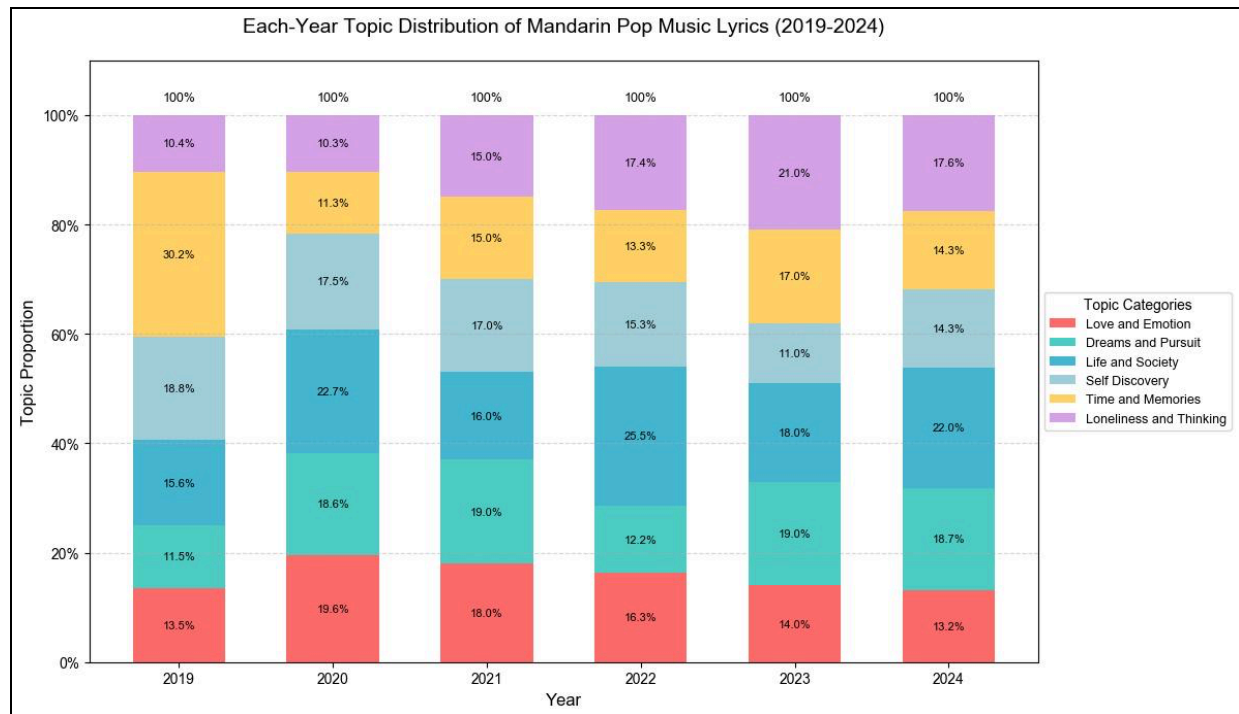


Figure 2: Each-Year Topic Distribution of Mandarin Pop Music Lyrics 2019-2024

The annual thematic priorities and relationships shift shows a noticeable decrease of the Time and Memories segment from 2019 to 2020, along with the expansion of the Love and Emotion and Life and Society segments, visually depicts a swift shift in lyrical focus driven by significant milestones such as the 70th anniversary of the founding of the People's Republic of China and the conclusion of the decade (Figure 2). The chart also reveals periods where topics may conflict or align; for instance, the simultaneous growth of Dreams and Pursuit and Life and Society around 2022-2024 suggests a dual emphasis on collective progress and personal aspirations. Overall, this graph highlights Mandopop's responsiveness to lyrical topics, showing how its focus continually evolves to reflect the changing cultural landscape.

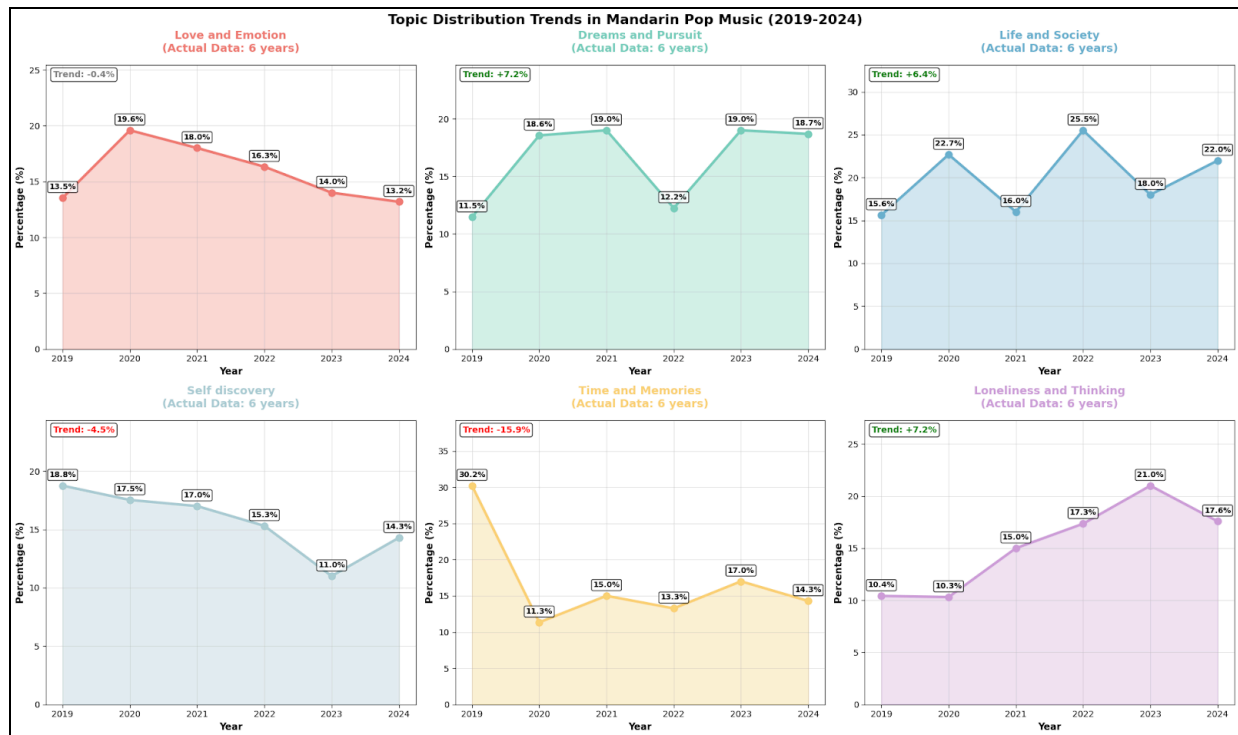


Figure 3: Topic Distribution Trends in Mandarin Pop Music Lyrics (2019-2024) by year

The trends above (Figure 3) disaggregate the trends, offering detailed trajectories for each topic.

- Love and Emotion.** The graph shows an overall decline of 0.4%, with a sharp peak in 2020 (19.6%) relative to 2019 (13.5%), followed by a gradual decline to 13.2% by 2024, displaying a progressive decline of romantic expressions.
- Dreams and Pursuit.** The graph shows a strong overall trend of 7.2%, beginning at 11.5% in 2019, steadily rising to 19.0% in 2021, dropping to 12.2% in 2022, and rebounding to 19.0% in 2023, presenting a refocus of aspiration.
- Life and Society.** The graph shows a strong overall trend of 6.4%, with peaks in 2020 (22.7%) and 2022 (25.5%), then dropping to 18.0% in 2023 and slightly recovering to 22.0% in 2024, displaying an oscillatory pattern of societal reflection.
- Self-Discovery.** The graph shows a 4.5% decline from 18.8% in 2019 to a low of 11.0% in 2023, followed by a slight recovery to 14.3% in 2024, indicating a shift away from deep introspection.
- Time and Memories.** The graph shows the most significant decrease of 15.9%, from 30.2% in 2019 to 11.3% in 2020, and fluctuating at lower levels thereafter, indicating a rapid shift away from nostalgia.
- Loneliness and Thinking.** The graph shows a 7.2% upward trend, remaining around 10% in 2019-2020, then increasing to 21.0% in 2023 before slightly decreasing to 17.6% in 2024, highlighting a growing focus on solitude and introspection.

The thematic landscape of Mandopop lyrics from 2019 to 2024 reflects a lively portrayal of Chinese society undergoing significant changes, especially those triggered by the COVID-19 pandemic. While the overall distribution (Figure 1) indicates a balanced focus across topics – with Life and Society the most

common (20.0%) and other topics closely grouped—the yearly trends (Figures 2 and 3) reveal a highly responsive and adaptable lyrical scene.

The onset of the COVID-19 pandemic in 2020 caused a significant shift in lyrical topics. The most noticeable change was the sharp drop in references to Time and Memories, from a dominant 30.2% in 2019 to just 11.3% in 2020 (Figure 3). This sudden shift away from nostalgic reflection, as shown by the shrinking yellow segment in Figure 2, indicates a collective move from focusing on the past to prioritizing immediate, existential concerns. At the same time, lyrics began to focus more on topics of human connection and shared experience. Love and Emotion increased to 19.6% in 2020, and Life and Society rose to 22.7% (Figure 3), showing a societal desire for closeness and a greater awareness of everyday realities during lockdowns.

During 2021 and 2022, as China continued its recovery efforts amid changing challenges, Mandopop lyrics also evolved. The enduring popularity of Dreams and Pursuit, which consistently accounted for a significant share of topics and reached 19.0% in both 2021 and 2023 (Figure 3), reflects a lyrical expression of national resilience and ongoing progress. The resurgence and peak of Life and Society at 25.5% in 2022 (Figure 3), overshadowing other topics in that year's bar chart (Figure 2), closely align with major collective events such as the Shanghai lockdown, underscoring how lyrics directly echo societal experiences. Notably, Self Discovery declined to 11.0% in 2023 (Figure 3), indicating a temporary shift away from personal introspection toward collective survival and external issues.

In 2023 and 2024, a clear shift was observed in the steady increase in topics related to Loneliness and Thinking. Although these topics initially accounted for a modest share (Figure 1), they showed consistent growth (Figure 3: +7.2% overall), reaching 21.0% in 2023. This suggests a deeper societal focus on introspection and solitude, moving beyond immediate crisis responses to address more profound psychological effects.

In essence, Mandopop's lyrical development during this period serves as a clear indicator of societal change. It transitioned from an early reactive stage, in which topics of connection and aspiration were prominent as coping mechanisms, to phases of confrontation with collective issues, and finally to a new phase of introspective reflection on personal psychological experiences amid rapid societal change.

4.3 Word Clouds Interpretation.

The annual word clouds (Figures 4-9) visually represent the most frequent words in the lyrics each year, providing qualitative insights into the evolving lyrical focus.



Figure 4: 2019 Lyrics Word Cloud



Figure 5: 2020 Lyrics Word Cloud



Figure 6: 2021 Lyrics Word Cloud



Figure 7: 2022 Lyrics Word Cloud



Figure 8: 2023 Lyrics Word Cloud



Figure 9: 2024 Lyrics Word Cloud

Table 3. High-frequency words & Relevant topics per year.

Year	High-frequency words	Relevant topics
2019	世界(world), 回忆(memory), 不会(won't), 时间(time), 希望(hope), 奉献(sacrifice)	Emphasize new environments, aspirations, and intimacy.
2020	世界(world), 时间(time), 想要(longing for), 遇见(encounter), 心跳(heartbeat)	Focus on <i>Love and Emotion</i> , rather than the past.
2021	世界(world), 不会(won't), 时间(time), 无人(nobody), 跳针(skip stitch), 拥抱(embrace)	Indicate a wish for optimism as the world begins to recover from the pandemic, the lyrics in this year embody hope and future thinking.
2022	世界(world), 不会(won't), 时间(time), 相遇(encounter), 拥抱(embrace), 温柔(gentle)	Reflect on resilience and reunion in the lyrics as pandemic restrictions are lifted, enabling people to reconnect with others.
2023	世界(world), 不会(won't), 时间(time), 想要(longing for), 孤独(lonely), 拥抱(embrace)	Heavy connections to <i>Loneliness and Thinking</i> , unlike previous years.
2024	世界(world), 我会(I will), 期待(hope), 是否(whether), 人生(life)	A contrasting shift from last year is present, from <i>Loneliness and Thinking</i> to certainty and anticipation of the future.



Figure 10: Combined Preprocessed Lyrics Word Cloud 2019-2024

Analysis of all the cloud datasets reveals that word frequency often highlights the main topics of a given year. For example, 2023 uniquely emphasizes topics like Loneliness and Thinking more than other years. It concerns the resumption of regular social interactions following the COVID-19 pandemic, a desire to reconnect and adjust to the new normal. Reasons for the sudden focus and annual shifts will be further explored in the sentiment analysis section, which investigates each topic's sentiment shifts and societal event relations.

Additionally, the most frequent words across all word clouds are 世界 (world) and 时间 (Time), reinforcing the identification of Life and Society and Time and Memories as the most frequently focused themes in the topic modeling section. Furthermore, words such as 奉献 (sacrifice), 孤独 (loneliness), and 人生 (life) are perceived as profound. It confirms the concept that Mandopop emphasizes collective human issues.

4.4 Sentiment Analysis.

4.4.1 Sentiment Analysis Overview.

Before the analysis, the four sentiments are defined as follows.

- (a) **Positive.** The “Happy/Romantic” Score: It defines a metric based on words in the “positive_words” set, including keywords such as 爱情 (Love), 幸福 (Happiness), 快乐 (Joy), 梦想 (Dream), 微笑 (Smile), and 阳光 (Sunshine). This score reflects the song's degree of optimism,

romance, or uplifting content; higher scores indicate that it is likely a sweet love song or an anthem about pursuing dreams.

- (b) **Negative.** The “Sad/Heartbreak” Score: It defines a metric based on words in the “negative_words” set, including keywords such as 分手 (Breakup), 眼泪 (Tears), 心痛 (Heartache), 寂寞 (Loneliness), 绝望 (Despair), 背叛 (Betrayal). This score reflects the song’s degree of tragedy, loss, or pain; higher scores indicate that it is likely a ballad about a breakup, depression, or regret.
- (c) **Ambiguous.** The “Nostalgic/Reflective” Score: It defines a metric based on words in the “ambiguous_words” set, including keywords such as 回忆 (Memories), 时间 (Time), 等待 (Waiting), 缘分 (Fate), 青春 (Youth), 岁月 (Years/ Age). This score reflects the song’s degree of sentimentality or nostalgia, not its inherent goodness or badness, as 回忆 (Memories) can be happy or sad, and 等待 (Waiting) can be romantic or painful.
- (d) **Neutral.** The “Filler/Descriptive” Score. It defines a metric based on words in the “Total Words - (Positive + Negative + Ambiguous)” set, including keywords such as 看见 (See), 走路 (Walk), 苹果 (Apple), 颜色 (Color). These words have passed the cleaning filter, but they do not convey a strong emotional charge. Most songs tend to have a high neutral score because lyrics generally include many descriptive words that aren’t emotional keywords.

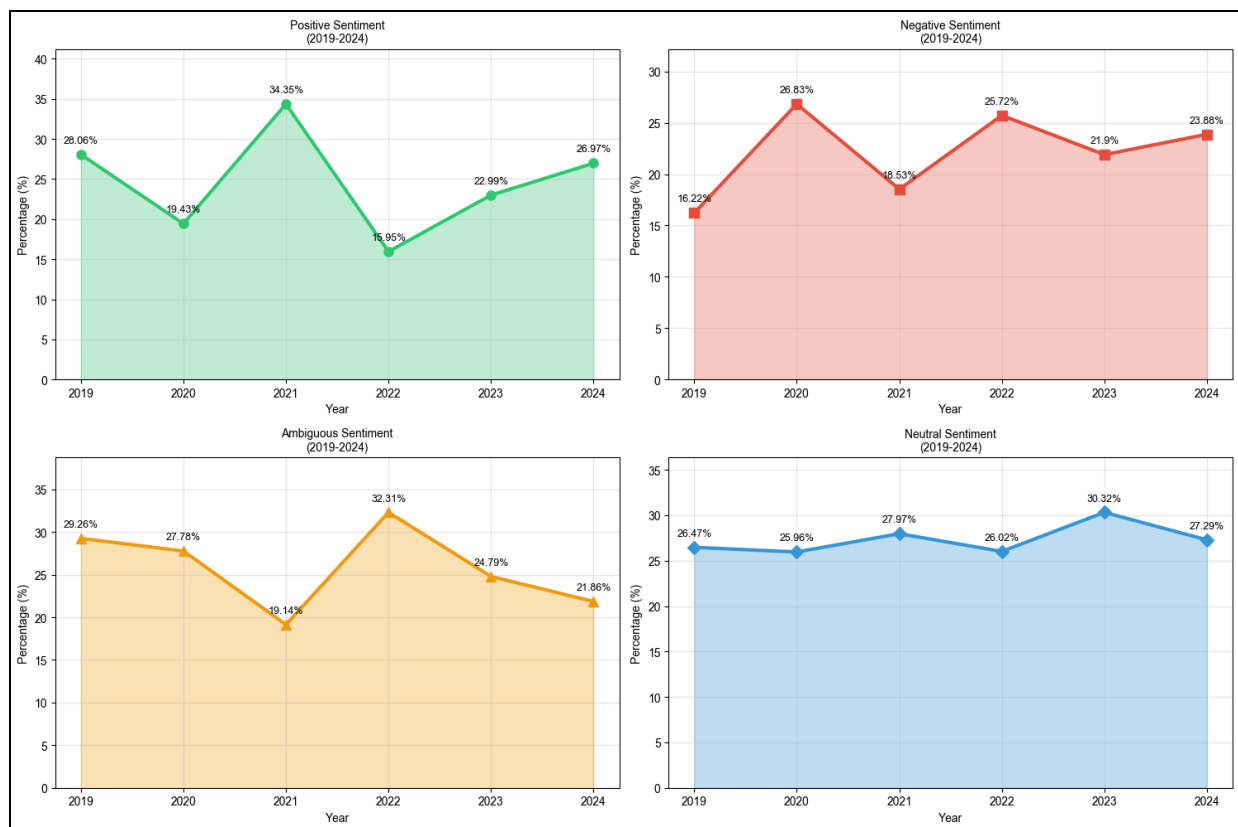


Figure 11: Sentiment distribution of Mandarin pop music lyrics from 2019 to 2024 by year

The sentiment analysis from 2019 to 2024 reveals a dynamic emotional landscape shaped by significant societal changes. In 2019, sentiments were predominantly positive, characterized by cultural confidence and rich emotional experiences among youth, especially in love and personal aspirations. This optimism

was overshadowed in 2020 by the COVID-19 pandemic, which led to a sharp spike in negative sentiment amid feelings of uncertainty and isolation. By 2021, a renewed sense of purpose and resilience emerged as society began to recover, leading to a significant rebound in positive sentiment. However, in 2022, there was a resurgence of negative and ambiguous feelings stemming from societal tensions and lockdown fatigue. In 2023, nuanced discussions of mental health emerged, reflecting struggles and pursuits of authenticity. Finally, in 2024, mixed sentiments emerged amid economic concerns and AI issues, alongside transformative expressions of love and self-healing, leading to a renewed sense of hope.

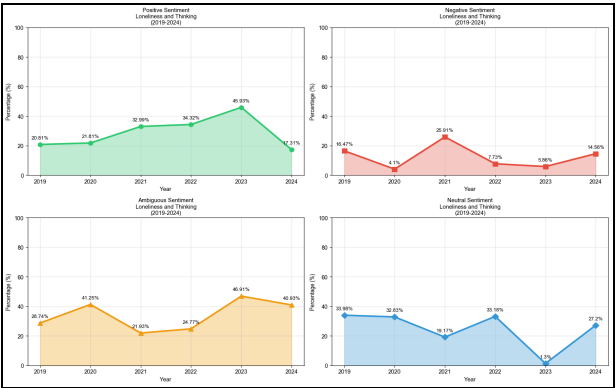


Figure 12: Sentiment distribution of Loneliness and Thinking in Mandarin pop music lyrics from 2019 to 2024 by year

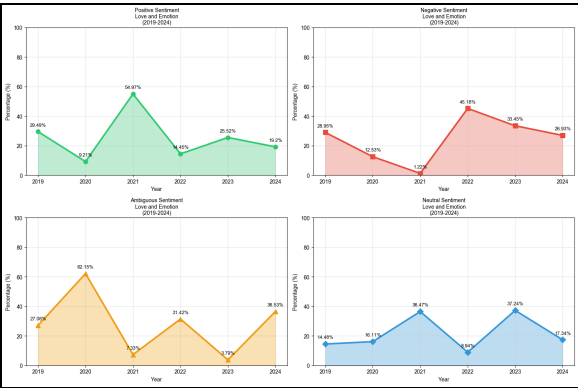


Figure 13: Sentiment distribution of Love and Emotion in Mandarin pop music lyrics from 2019 to 2024 by year

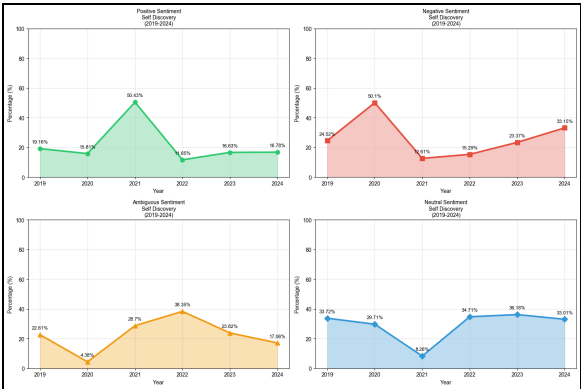


Figure 14: Sentiment distribution of Self Discovery in Mandarin pop music lyrics from 2019 to 2024 by year

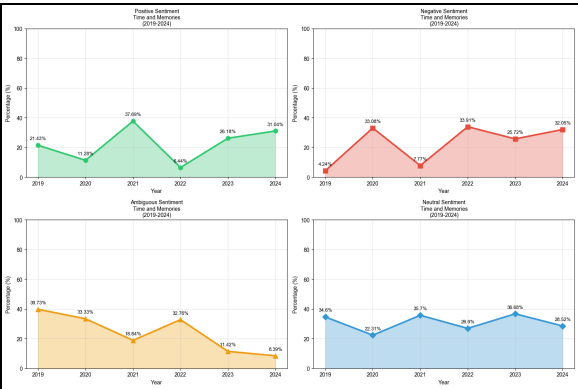


Figure 15: Sentiment distribution of Time and Memories in Mandarin pop music lyrics from 2019 to 2024 by year

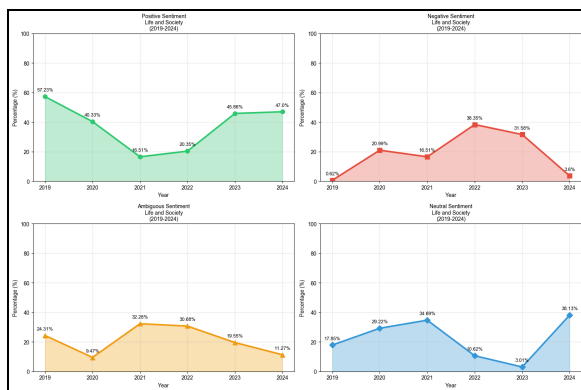


Figure 16: Sentiment distribution of Life and Society in Mandarin pop music lyrics from 2019 to 2024 by year

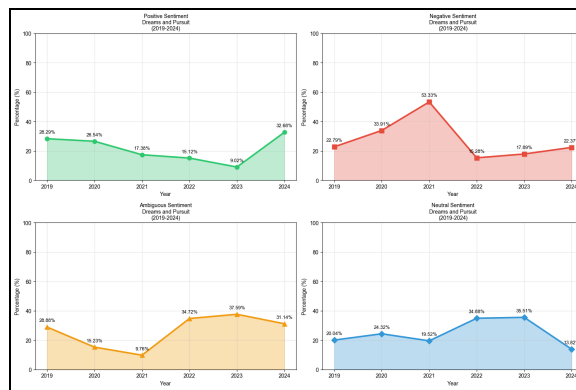


Figure 17: Sentiment distribution of Dreams and Pursuit in Mandarin pop music lyrics from 2019 to 2024 by year

4.4.2 Sentiment Analysis on Six Thematic Trends.

(a) Loneliness and Thinking.

Key fluctuations indicate that positive sentiment peaked at 45.93% in 2023 and declined to 17.31% in 2024. Negative sentiment rose to 25.91% in 2021, then decreased to 5.86% in 2023, and subsequently increased to 14.56% in 2024.

In 2021, peak negative sentiment reached 25.91%, as the recovery phase masked lingering pandemic trauma. While young people resumed their routines, many still felt disconnected, experiencing social awkwardness after prolonged isolation. The tension between a longing for normalcy and unaddressed loneliness fueled negative feelings. By 2023, positive sentiment peaked at 45.93%, while ambiguous sentiment reached 46.91%. The deaths of Coco Lee and Dao Lang’s satirical album heightened public awareness of mental health issues. Loneliness was reframed as a shared experience, exemplified by discussions of emotional vulnerability online. In 2024, positive sentiment declined as economic anxieties—such as unemployment and AI disruptions—dominated public concern. Loneliness transitioned from a reflective experience to a source of dismay. The focus on survival left little room for constructive engagement with loneliness, resulting in a decline in positive sentiment.

(b) Love and Emotion.

Key fluctuations indicate that positive sentiment rose from 9.21% in 2020 to 54.97% in 2021, then declined to 19.2% in 2024. Negative sentiment, on the other hand, hit a low of 1.22% in 2021 before surging to 45.18% in 2022.

In 2020, during the peak of ambiguous sentiment at 62.15%, pandemic-related physical barriers, such as quarantine and social distancing, heightened the urgency of emotional connection but left it unfulfilled. Lyrics centered on warmth and hugs, yet the crisis’s uncertainty left emotions unresolved, resulting in a surge of ambiguous sentiment. By 2021, positive sentiment peaked at 54.97%. After the isolation of 2020, there was a resurgence of offline gatherings and reunions. Topics such as companionship, exemplified by songs like “Empty World (这世界那么多人)”, and patriotic family love, as seen in “As You Wish (如愿)”, resonated widely. The shift from survival to recovery heightened longing for emotional bonds, pushing positive sentiment to its highest point in six years. In 2024, negative sentiment rose to 26.93%, driven by societal tensions, such as gender conflicts depicted in “Lost in the Stars (消失的她)” and its theme song, “Snare (笼)”. Additionally, post-pandemic fatigue from “compensatory love” shifted focus to emotional

constraints, including control within relationships. Coupled with economic anxiety, these factors increased negative feelings surrounding love and intimacy.

(c) Self Discovery.

Key fluctuations indicate that positive sentiment peaked at 50.43% in 2021 and subsequently declined to 16.78% in 2024. Conversely, negative sentiment spiked to 50.1% in 2020, then decreased to 12.61% in 2021 before rising again to 33.15% in 2024.

In 2020, negative sentiment peaked at 50.1%, primarily driven by the pandemic's impact, which caused widespread personal crises such as unemployment, pay cuts, and academic pressure. Young people particularly faced identity disruptions through career changes and social isolation, leading to increased self-doubt and a high level of negative feelings regarding self-discovery. By 2021, positive sentiment peaked at 50.43%, driven by regulations in the entertainment industry aimed at reforming fan culture. This shift encouraged audiences to question idealized images of idols and to embrace authenticity, exemplified by songs such as "A Cool Fish (无名之辈)", which celebrated ordinary individuals and fostered self-acceptance. However, by 2024, negative sentiment had risen again to 33.15%, driven by a wave of unemployment and anxiety surrounding AI-driven job displacement. Young people are experiencing rootlessness – both geographic and psychological – and economic stagnation, which has heightened negative feelings about self-worth and personal direction.

(d) Dream and Pursuit.

Key fluctuations indicate that positive sentiment has decreased from 2019 to 2023, before peaking in 2024 at its highest level over six years. Negative sentiment spiked in 2021 to 53.33% and rebounded slightly in 2023 and 2024.

In 2021, talent shows – powerful platforms that encourage young people to pursue their dreams – were comprehensively banned following incidents of blind fanaticism that diminished positive expressions of dreams. Conversely, negative emotions peaked at 53.33% in 2021. The outbreak of involution discourse and frequent celebrity misconduct incidents, such as those involving Kris Wu and Li Yifeng, led young people to become resistant to pursuing dreams for utilitarian reasons, and the music scene was filled with anxiety about dreams. By 2023, under the topic of Dreams and Pursuit, positive emotions declined from 2019 to 2023. This was mainly due to the pandemic, which brought society to a standstill, forced people into isolation, and made it more challenging to chase goals. In 2024, the recovery of the offline performance market revitalized this topic. Many old songs regained popularity, and programs like "Call Me By Fire" rekindled middle-aged people's passion for pursuing dreams.

(e) Life and Society.

Key fluctuations indicate that positive sentiment peaked in 2019 at 57.23%, declined to a low of 16.51% in 2021, and then rebounded from 2022 to 2024. Negative sentiment, on the other hand, started at 0.62% in 2019, peaked at 38.35% in 2022, and then declined sharply to 3.6% in 2024.

In 2019, within the Life and Society topic, positive emotions peaked at 57.23% at the start of the year. This coincided with a surge in mainstream themes, driven by the 70th anniversary of the founding of the People's Republic of China, which also inspired songs like "My People, My Country" and "The Youth's China". In 2021, influenced by the pandemic, public social emotions gradually shifted toward anxiety. Combined with social pressure from rising unemployment and the rise of AI, positive emotions continued to decline, reaching a low point in 2021. In 2022, negative emotions surged to 38.35%, reflecting a series of adverse societal events, including pandemic lockdowns, economic downturns, and high-profile tax evasion cases involving celebrities such as Deng Lun and Viya. Frustration with current realities and societal issues also

surfaced in song lyrics, as in “Mohe Ballroom”.

(f) Time and Memories.

Key fluctuation indicates that positive sentiment peaked in 2021 at 37.69%, dipped to a low of 6.44% in 2022, then rebounded in 2023-2024. Negative sentiment began at 4.24% in 2019, fluctuated between 2019 and 2022, and remained at a mid-high level during 2022-2024.

In 2021, during the topic of Time and Memories, positive emotions peaked at 37.69%, coinciding with the first year after the pandemic and the surge of nostalgic shows like “Sisters Who Make Waves”. People longed for pre-pandemic times, with songs like Time Machine evoking sweet memories and nostalgia. Conversely, 2020 and 2022 saw peaks in negative emotions, corresponding to the two major COVID-19 waves in China. During these times of anxiety and tension, people yearned for freedom and experienced regret, with song lyrics often expressing anxiety about the passage of time.

5. Limitations and Future Studies.

First, while TME rankings provide insights into popularity, they do not offer a full view of the Mandopop scene; as a result, they overlook niche genres that may not receive mainstream exposure.

Second, the granularity of the Latent Dirichlet Allocation (LDA) topic modeling may obscure overlapping themes common in poetic Mandarin lyrics. We address this by linking songs to a single thematic topic based on the calculated relevance coefficient, even though there may be some overlap or imperfect categorization.

Third, sentiment lexicons pose challenges because they often miss the metaphor-rich expressions in song lyrics. Dictionary-based sentiment analysis thus struggles with context, making it hard to detect words whose sentiment depends on surrounding text or specific domains, especially nuances such as sarcasm, irony, and metaphor (Kharde & Sonawane, 2016). These methods generally have lower accuracy than machine learning models, which can learn complex patterns from data. We address this by adding extra keywords to reduce these issues.

Overall, these limitations underscore the need for more sophisticated analytical approaches to better capture the artistic and emotional depth of Mandopop lyrics.

Future research should therefore incorporate contextual embeddings to reevaluate the accuracy of the results, for example, via BERT-based topic clustering. Notably, unlike lexicon-based approaches, pre-trained transformer models such as BERT have demonstrated exceptional contextual comprehension by capturing subtle differences in word meanings (e.g., distinguishing “sharp” in “sharp mind” from “sharp decline”) (Devlin et al., 2019). They also achieve state-of-the-art results across various NLP benchmarks because of their deep architecture and extensive pre-training on large datasets (Devlin et al., 2019). In addition to incorporating the BERT model, future research could analyze social media commentary and compare music markets in Mainland China with those in Taiwan, Hong Kong, or Southeast Asia to develop a broader, more comprehensive understanding of the relationship between Mandopop and societal shifts.

6. Conclusion.

In all, Mandopop from 2019 to 2024 serves as a cultural mirror, explicitly reflecting China's societal changes each year. They successfully capture thematic representations, common topics, and insights into China's society, including both historical events and the public's emotional experiences. Topic modeling reveals diverse lyrical themes, evolving from connection and aspiration in early stages to confrontation with societal issues and, later, introspection into personal psychological experiences amid rapid change. Sentiment analysis indicates a transition from optimism in 2019 through pandemic-related challenges to a complex emotional state characterized by resilience, love, and self-discovery. These themes mirror societal moods and tensions, highlighting Mandopop's role as a cultural barometer that fosters connection and understanding during societal transformation.

AI Usage Disclosure

Codes in Jupyter notebooks have been generated and adjusted by AI. Manual edits and human reviews have been performed to ensure code accuracy and the quality of output graphs. The AI utilized is ChatGPT.

Contribution²

Introduction & Significance: Jimmy, Sophie

Literature Review: Jimmy, Sophie

Methodology and Data: Audrey, Jimmy, Leo

Findings and Analysis (Topic Modeling): Jocelyn, Leo, Yuki

Findings and Analysis (Word Cloud): Audrey, Jocelyn, Leo, Yuki

Findings and Analysis (Sentiment Analysis): Jimmy, Jocelyn, Leo

Limitations and Future Studies: Jimmy, Yuki

Conclusion: Jimmy, Yuki

Code Design: Jimmy, Yuki (with ChatGPT assistance)

Poster Design: Jimmy, Yuki

Poster Printing: Jocelyn, Leo

Raw Data Processing: All

All in Revision: Jimmy, Jocelyn, Yuki

² Listing names in alphabetical order by English names.

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